



Centre Monétique Interbancaire

Online payment integration

Version 2.0

Version History

Version N°	Version Date	Type of change
V1.0	05 November 2015	Creation
V1.1	19 November 2016	« encoding » parameter with « utf-8 » value
V1.2	02 February 2017	« shopurl » parameter
V1.3	02 March 2017	« currenciesList », « amountCur », « symbolCur » parameters
V1.3.1	13 April 2017	BUSINESSDATE parameter is no more used
V1.3.2	18 August 2017	"autoRedirect" parameter
V.1.4	06 August 2018	Errors codes list + Transactions statuses + Callbacks control and monitoring
V1.4.1	12 December 2018	"AutoRedirect" parameter name update
V1.4.2	14 February 2019	"BillToCountry" parameter format "BillToName" and "email" parameters are mandatory - New "sessiontimeout" parameter - Hash calculation in the case of items containing the word "document" - ISO8583-13 error code added to errors list
V1.4.3	13 May 2019	New merchant's response "FAILURE" to CMI callback
V1.4.4	14 January 2020	"description" parameter's length updated. - Errors codes list updated. - Notice in regards to hash calculation.
V2.0	22 March 2022	

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1. INTRODUCTION

This document gives a detail description of the technical specifications to connect a merchant web site to the online payment gateway of the Centre Monétique Interbancaire (CMI).

2. SECURITY OF EXCHANGES

For securing the online payment transactions, CMI uses SSL protocol to encrypt the merchant's order information, customer's payment information and merchant's screens in the payment platform used for transactions monitoring. For this, CMI uses SSL certificates signed by trusted certificate authorities.

Security is assured at several levels.

- On the merchant's web site, every payment request is sent with a hash code which is generated with a hash algorithm and a secret key. This key is generated by the merchant and configured in its administrative interface of the payment platform.
- On the platform, the same hash algorithm is used with the merchant's shared secret key to authenticate the merchant's web site and to verify that payment request's elements have not been altered during their transfer across the network.
- Filling the information required for the financial processing is secured by SSL protocol with other algorithms which are not described in this document.

Cryptographic functions are not accessible to the user and thus cannot be amended or modified.

3. ON LINE PAYMENT PROCESS BY CREDIT CARD

3.1. Steps :

The process of online payment by credit card takes place according to the following steps:

- The client generates its order in the merchant site and chooses to pay online by credit card.
- The customer is redirected to the CMI platform that displays the payment screen. On this page is displayed a summary of the customer's order with the transaction's amount. The customer is thus invited to fill its payment credentials (payment card's number and expiration date and CVS code).
- When the client validates his payment form, the payment platform checks if the payment card used is authenticable. If it is the case, the customer is redirected to an authentication screen where he must fill his authentication secret code that is verified by its bank.
- In the case of a successful authentication, the payment platform sends an authorization request to the acquirer (CMI) that treats it with the issuer.
- If payment authorization is rejected (insufficient funds, the authorized card's ceiling exceeded, blacklisted card,...), an error message is displayed on the customer's screen. The page allows the client to retry payment.

Notice: The error message displayed to the client does not necessarily give the exact reason for the rejection. This is conform to the standards of fraud prevention. If the customer needs to know the reason for the payment failure, he is supposed to contact his bank. The merchant has access to the error message via its CMI back office of online transactions monitoring.

- If the payment authorization is granted by the customer's Bank
 - If the merchant's web site has asked it in the initial payment request, a confirmation request for payment is sent automatically in the background (server-to-server) from CMI platform to the merchant's web site.
 - When receiving the payment confirmation's request, the merchant site identifies the concerned order in its database, updates its status and register all the payment information contained in the request. It also returns an acknowledgement response to the CMI platform. During this exchange, the merchant's web site can ask the CMI platform to confirm the transaction in order to debit the customer or to cancel the transaction in order to unlock the transaction's amount in the customer's account.
 - In the CMI platform, an online receipt is displayed to the customer containing the order and the payment information (Order ID, Payment ID, payment date, payment method ...). This receipt can include a link to allow the client go back to the merchant's website to finish the purchasing process.

Notice: When a bank accept a payment authorization, it applies a lock of the transaction's amount on the customer's account. It is a payment guaranty that can take 7 days. The amount is unlocked in the customer's account in the following cases:

- When the transaction is confirmed (and the account is debited).
- When the merchant cancels the transaction on the same day of its authorization.
- If no settlement of the transaction is sent to the bank within the payment guaranty period (7 days)



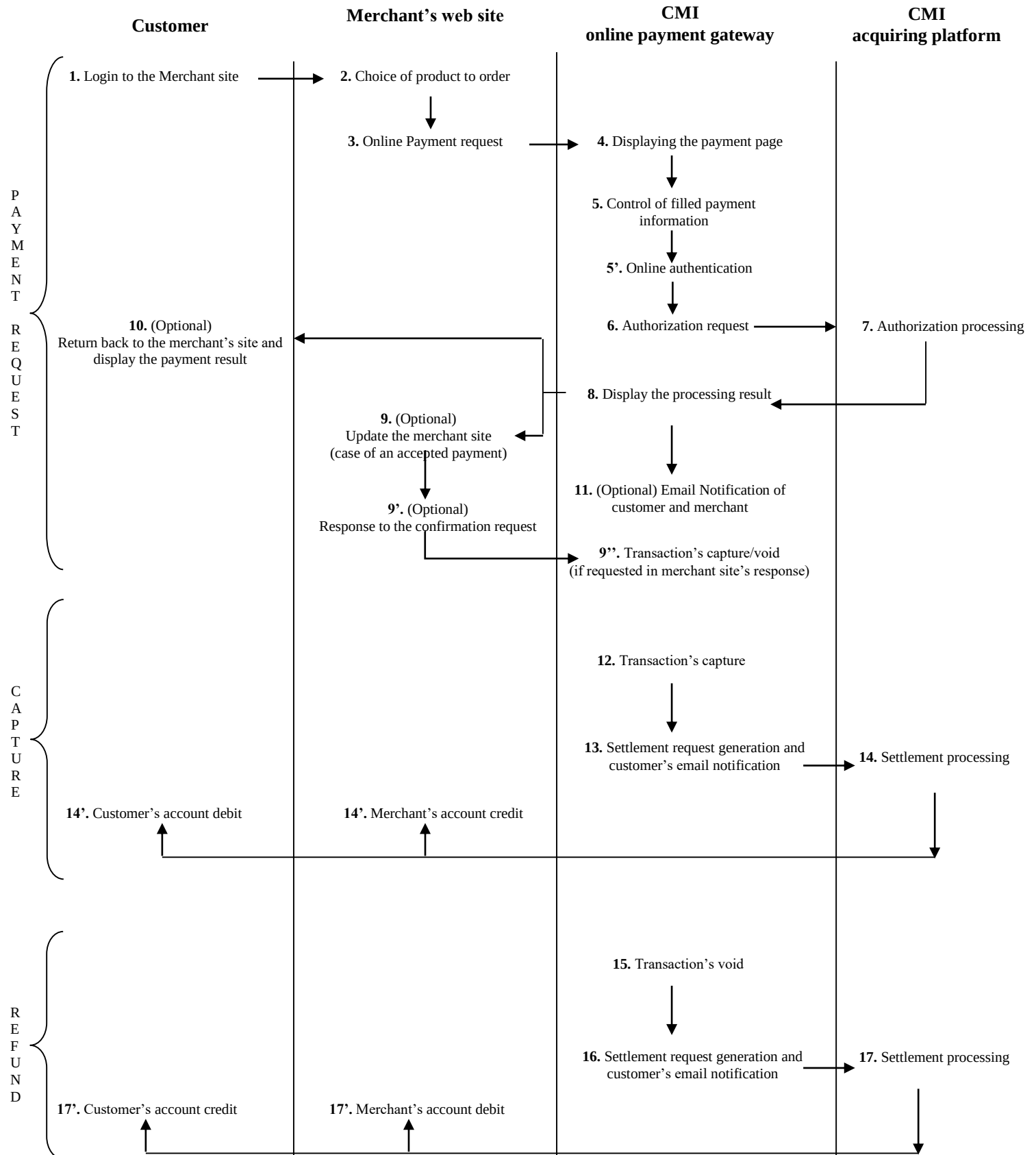
- In the case of confirmed transaction (manually via the merchant back office of the platform or automatically via the server-to-server payment confirmation request or via API), the transaction's settlement is sent by the payment platform to the acquirer to debit the customer's account and credit the merchant's account.

Notice: The merchant's account is credited with the total amount of all the transactions settled by the CMI during the day. CMI's treatment fees are deducted at source from the total amount.

- The merchant refund the amount of a transaction to the customer via the back office of the payment platform or via API.
- Notifications are sent by the CMI platform via emails to both customers and merchants.

Notice: The CMI platform notifications are enabled by default, but they can be disabled if the merchant wants to .

3.2. Flow Diagram:



4. EXCHANGED DATA

The data exchanged between the merchant's web site and the CMI's platform is sent through an HTTP form (form post). This exchange is secured using a hash algorithm and a secret key.

4.1. PAYMENT REQUEST:

Payment request form sent from the merchant's web site to the CMI's platform when redirecting the customer for the online payment.

4.1.1. Parameters :

This section provides a description of the necessary attributes for the creation of the payment request. These attributes are generally organized into the following categories:

- Mandatory attributes
- Optionel attributes

Mandatory attributes

Attribute Name	Description	Format	Constraint
clientid	Merchant's id assigned by CMI	Alphanumeric (15)	Required
storetype	Merchant payment model	Alphanumérique (15)	Required Value to use for Hosting Payment Page : « 3d_pay_hosting »
trantype	Transaction type	Alphanumérique (15)	Required Set to "PreAuth" for preauthorization
amount	Ttransaction amount	Numeric Amount value without currency symbol. Use "." or "," as decimal separator. Example: 29.95	Required
currency	ISO code of the transaction currency	Numeric(3)	Required ISO 4217 numeric currency code. ISO code for MAD: 504
oid	Unique Order/Cart identifier in the merchant site's database	Alphanumeric (64)	Required

okUrl	The URL used to redirect the customer back to the merchant's web site in case of accepted payment authorization.	URL	Required
failUrl	The URL used to redirect the customer back to the merchant's web site in case of failed/rejected payment authorization.	URL	Required
lang	The language used to display the payment screens.	Alphabetic (2)	Required Possible values : • ar: Arabic • fr: French • en: English Default « fr »
email	Customer's e-mail	Alphanumeric (64)	Required Email Format
BillToName	Customer's name (first name and last name)	Alphabetic (255)	Required
rnd	Random string, will be used for hash comparison	Alphabetic (20)	Required
hash	Control hash code	Alphanumeric	Required (See below how to generate the hash code)
hashAlgorithm	Hash version	Alphanumeric	Required Value to use: "ver3"

Optional attributes

Attribute Name	Description	Format	Constraint
encoding	Encoding of the posted data	Alphanumeric (32)	Optional Possible value : utf-8
description	Description sent to MPI	Alphanumeric (125)	Optional
tel	Customer's phone number	Alphanumeric (32)	Optional
BillToStreet1	Customer's postal adresse 1	Alphanumeric (255)	Optional
BillToStreet2	Customer's postal adresse 2	Alphanumeric (255)	Optional
BillToCity	Customer's city	Alphanumeric (64)	Optional
BillToStateProv	Customer's state or province	Alphanumeric (32)	Optional

BillToPostalCode	Customer's postal code	Alphanumeric (32)	Optional
BillToCountry	Customer's country	Alphanumeric (3)	Optional
idN	Id of item #N, required for item #N	Alphanumeric (128)	Optional When there's one item, N is set to 1. In this case, the parameter name is : id1.
itemnumberN	Item number of item #N	Alphanumeric (128)	Optional
productcodeN	Product code of item #N	Alphanumeric (64)	Optional
qtyN	Quantity of item #N	Numeric	Optional
descN	Description of item #N	Alphanumeric (128)	Optional
priceN	Price of item #N	Numeric Amount value without currency symbol. Use "." or "," as decimal separator. Example: 29.95	Optional
totalN	Subtotal of item #N	Numeric Amount value without currency symbol. Use "." or "," as decimal separator. Example: 29.95	Optional
CallbackResponse	Enables/Disables the host-to-host callback request	Boolean	Optional Possible values : • true: Enabled • false: Disabled Default « true »
CallbackURL	The URL used for the host-to-host callback request	URL	Optional
shopurl	The return URL to which customer is redirected when he clicks on the button "Cancel" displayed in the payment page.	URL	Optional
currenciesList	Used to show (or not) the exchange currencies' list in the payment pages. This list is generated by the CMI ECOM platform. In this case, the merchant doesn't manage the values listed. If the merchant wishes to show, in the payment page, an exchange value calculated by its web site, he should use the parameters "amountCur" and "symbolCur".	Boolean	Optional Possible values : • true: Enabled • false: Disabled Default « false »

amountCur	Amount exchange (in a foreign currency) to show to the customer in the payment page. In this case, the value of the exchange amount is calculated by the merchant web site (unlike the case when the "currenciesList" parameter is used) .	Numeric Amount value without currency symbol. Use "." or "," as decimal separator. Example: 29.95	Optional
symbolCur	Currency symbol to be displayed in the payment page with the "amountCur" parameter	Alphanumeric	Optional Examples : EUR, USD.
AutoRedirect	Used in order to redirect the customer automatically back to the merchant's web site when the transaction is accepted.	Boolean	Optional Possible values : • true: Enabled • false: Disabled Default « false »
sessiontimeout	Allow to set a session time-out duration for the payment page.	Numeric The parameter is calculated in seconds.	Optional Example : 600 (for a session of 10 minutes) Default: 1800 seconds. Minimum allowed value is 30 seconds and the maximum is 2700 seconds.
redirectGet	Used to change the method of the request sent by the platform to the okUrl and failUrl URLs from POST to GET.	Boolean	Optional Possible values : • true: Enabled • false: Disabled Default « false »

Notices:

- It is recommended that the order's status update in the merchant's web site uses the server-to-server payment confirmation request via the CallbackURL parameter. The parameters okUrl and failUrl can also be used for this purpose, but they require customers to be redirected back (html) to the merchant's web site.
- The URL set in the parameter CallbackURL must be reachable on Internet even in testing phase.
- The URL set in the CallbackURL parameter should not require authentication (login/password).
- It is strongly recommended to use the "encoding" parameter, even if it is optional, to avoid problems when the data contains special characters.
- Care must be taken to respect the size of each parameter of the payment request.
- The string set in the oid parameter should not contain any special characters (° or € or parentheses, etc.)
- The "total" parameter should not be used in the payment request.

4.1.2.Payment request Example

```
<form method="post" action="https://host/fim/est3dgate ">
  <input type="hidden" name="clientid" value="9900000000000001"/>
  <input type="hidden" name="storetype" value="3d_pay_hosting" />
  <input type="hidden" name="hash" value="iej6cPOjDd4IKqXWQEznXWqLzLI=" />
  <input type="hidden" name="trantype" value="PreAuth" />
  <input type="hidden" name="amount" value="31.50" />
  <input type="hidden" name="currency" value="504" />
  <input type="hidden" name="oid" value="1291899411421" />
  <input type="hidden" name="okUrl" value="https://www.teststore.ma/success.php" />
  <input type="hidden" name="failUrl" value="https://www.teststore.ma/fail.php" />
  <input type="hidden" name="lang" value="en" />
  <input type="hidden" name="rnd" value="asdf" />
  <input type="hidden" name="hashAlgorithm" value="ver3">
```

Optional parameters

```
<input type="hidden" name="tel" value="012345678">
<input type="hidden" name="email" value="test@test.ma">
```

<!-- Billing Parameters [All Optional]-->

```
<input type="hidden" name="BillToCompany" value="Billing Company">
<input type="hidden" name="BillToName" value="Bill John Doe">
<input type="hidden" name="BillToStreet1" value="Address line 1">
<input type="hidden" name="BillToStreet2" value="Address line 2">
<input type="hidden" name="BillToCity" value="Casablanca">
<input type="hidden" name="BillToStateProv" value=" Casablanca">
<input type="hidden" name="BillToPostalCode" value="12345">
<input type="hidden" name="BillToCountry" value="504">
```

<!-- Order Item Parameters [All Optional]-->

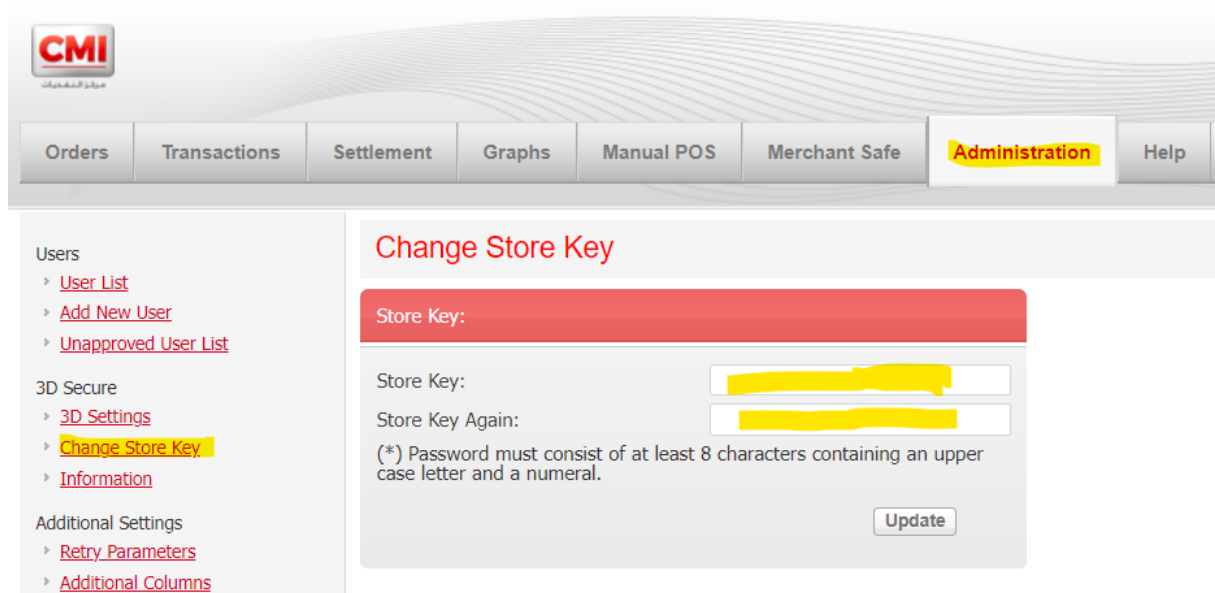
```
<input type="hidden" name="ItemNumber1" value="a5">
<input type="hidden" name="ProductCode1" value="a5">
<input type="hidden" name="Qty1" value="3">
<input type="hidden" name="Desc1" value="a5 desc">
<input type="hidden" name="Id1" value="a5">
```

```
<input type="hidden" name="Price1" value="10.50">
<input type="hidden" name="Total1" value="31.50">

</form>
```

4.1.3. Creating a hash key (storeKey)

To enter a hash key, you must connect to the CMI transaction monitoring Backoffice, then use the Administration -> Change store keys menu.



Notice :

- The hash key must not contain the word "SKS".

4.1.4. Hash generation of the payment request

Hashing Approach is used in order to authenticate users for transaction requests. To generate the hash for client authentication, append all posted request parameter values in alphabetical order (A to Z) using pipeline "|" as the separator between parameters. If a parameter is sent to CMI platform with an empty value, it still will be added to hash data (For an empty value example, email parameter value can be checked in below example).

After all request parameters are added alphabetically using pipeline "|" as separator, storeKey will be added to hash data again using pipeline "|".

Note that in a case of using “|” character in a parameter, “\” character is used for escaping. Therefore, if the “\” character is also used in a parameter as its own value, we use “\” character before it and then append it to hash plain text. For better understanding, you can check the example below :

Original Value : ORDER-256712jbs\j6b|
Value used for Hash Calculation : ORDER-256712jbs\\j6b\|

Example Parameters and Hash Calculation:

clientId	100200127
amount	95.93
okurl	http://localhost:8080/SampleCodeJSPTTest/GenericVer3ResponseHandler
failUrl	http://localhost:8080/SampleCodeJSPTTest/GenericVer3ResponseHandler
TranType	PreAuth
email	
callbackUrl	http://localhost:8080/SampleCodeJSPTTest/GateResponseControl.jsp
currency	504
rnd	87954458746
storeType	3d_pay_hosting
lang	en
hashAlgorithm	ver3
BillToName	name
BillToCompany	billToCompany
storeKey	ABCD1234

Hash:

Order of Used Parameters in Hash Data :

amount|BillToCompany|BillToName|callbackUrl|clientId|currency|email|failUrl|hashAlgorithm|lang|okurl|rnd|storetype|TranType|storeKey

Plaintext:

95.93|billToCompany|name|http://localhost:8080/SampleCodeJSPTTest/GateResponseControl.jsp|100200127|504||http://localhost:8080/SampleCodeJSPTTest/GenericVer3ResponseHandler|ver3|en|http://localhost:8080/SampleCodeJSPTTest/GenericVer3ResponseHandler|87954458746|3d_pay_hosting|PreAuth|ABCD1234

Hash = Hash =base64_encode (pack('H*',hash('sha512',plaintext)));

Notice:

- Parameters named "encoding" and "hash" are ignored during hash calculation.
- For security reasons, the CMI platform uses a tool that detects, prevents and disables the execution of unauthorized scripts (code contained in the HTTP parameters). So you have to replace any character after the string "document" with a dot "." during the calculation of the hash.
Examples:
document abc -> document.abc
documentabc -> document.bc
- In order to avoid a hash error, you must ensure that the values of the calculated parameters are the same as those sent in payment request.
For example, adding a submit button that will perform the action of sending the payment form may influence the hash calculation. To remedy this case, you must calculate the value of this element (button) in the hash calculation or delete its attribute "name".
- In order to avoid a hash error, it must be ensured that the website correctly calculates the hash in the event of the use of spaces at the end and at the beginning of the string in the personal parameters of the customer. Otherwise, these spaces must be deleted from all the parameters sent in the payment request.

4.2. PAYMENT RESULT :

When the customer submits the payment page with his card credentials, the 3D authentication flow starts. The payment authorization is then been processed with the customer's issuer bank. Payment response parameters and some or all parameters that were sent by the merchant's web site will be posted back to the merchant's web site by the CMI platform. In case of an accepted payment, the merchant's web site updates the order status in its DB and can ask the CMI platform to confirm the payment for the customer to be debited.

4.2.1. Host-to-host callback:

The merchant starts payment flow by sending a PreAuth request to CMI platform with the parameter CallbackResponse set to the value "true" (CallbackResponse=true). In addition to the required parameters, CallbackUrl should be sent so that CMI platform can send the confirmation callback to the merchant.

What should be noticed for this callback request in server-to-server mode, is that it reminds the merchant's web site as well of the cases of rejections as cases of successful transactions. So it may well be that the merchant's web site receives, for the same transaction (same order number), rejection returns that will be followed by a return of transaction acceptance. This means that the client has failed attempts before it succeeds. It is the parameter ProcReturnCode which makes it possible to distinguish between callback requests of successful payment authorizations (ProcReturnCode = 00) and the callback requests of the payment failures (ProcReturnCode! = 00 or nonexistent).

Merchant's web site response:

Upon receiving the CMI callback request, the merchant's web site is expected to send callback response to CMI platform. The following callback responses are supported:

- **Acknowledgement:** Merchant's response should be "APPROVED".
The transaction is acknowledged by the merchant but he asks to not debit the customer. The merchant needs to manage capture or void manually via CMI Merchant Center interface.
- **Acknowledgement&Settle:** Merchant's response should be "ACTION=POSTAUTH".
In this case, an automatic capture (PostAuth) request will be sent to the acquirer bank for the customer to be debited.
- **Failure:** Merchant's response should be "FAILURE".
The merchant website fails to take into account the transaction. The customer is asked to check the status of his/her order on the merchant's site. The merchant needs to manage capture or void manually via CMI Merchant Center interface.
- **Timeout:** If timeout occurs with the merchant's web site, the customer is asked to check the status of his/her order on the merchant's web site. Merchant needs to manage capture or void manually via Merchant Center interface.
- **Merchant's response syntax error:** If there is an error from the merchant's web site, the customer is asked to check the status of his/her order in the merchant's web site. Merchant needs to manage capture or void manually via Merchant Center interface.

Merchant's web site controls:

During the callback request, the merchant's web site is supposed to do the following:

- Generate a hash code with the same data posted by the CMI platform in the callback request. Then compare this calculated hash with the hash sent by the CMI platform in the callback request. If they are identical, proceed to the next check.
- Look, in the orders' DB of the merchant's web site, for the record identified by the value of the "oid" parameter sent by the CMI platform in the callback request.
- Check if the amount of the order recorded in the orders' DB of the merchant's web site is equal to the amount sent by the CMI in the callback request via the "amount" parameter.
- Check the "ProcReturnCode" parameter value sent by the CMI in the callback request:
 - If ProcReturnCode = 00, this is an accepted transaction.
 - So it is necessary to update the status of the order in the orders' DB of the merchant's web site (status = Paid).
 - Answer the CMI callback request with:
 - "ACTION=POSTAUTH": in order to debit the client automatically.
 - "APPROVED": in order to not debit the client automatically. In this case, the merchant needs to manage capture or void manually via CMI Merchant Center interface.
 - If the ProcReturnCode $\neq$ 00 or if ProcReturnCode parameter does not exist in the callback request, it is a payment authorization failure.
 - In this case, do not change the status of the order in the BDD orders of the merchant's web site.

- The response to return to the CMI callback request is "APPROVED" (acknowledgment).
- If a technical problem occurs in one of the previous steps, answer the CMI callback request with "FAILURE". In this case, the merchant needs to manage capture or void manually via CMI Merchant Center interface.

Notice:

- *The callback request sent by CMI platform to the merchant's web site in server-to-server mode, reminds as well of the cases of successful transactions as cases of rejections. So it may well be that the merchant's web site receives, for the same transaction (same order number), rejection returns that will be followed by a return of transaction acceptance. This means that the client has failed attempts before it succeeds. It is the parameter ProcReturnCode which makes possible to distinguish between the callback request of a successful payment authorization (ProcReturnCode = 00) and the callback request of a failed payment (ProcReturnCode! = 00 or nonexistent).*

Callback failure :

This section describes the appropriate way to handle payment cases via APIs where the transaction confirmation server-to-server callback fails.

Indeed, when the customer returns via the okURL to the merchant's website after payment, the website must check whether a payment confirmation callback has previously been received from the CMI platform:

- If the callback was received, display a payment confirmation message to the customer.
- If no callback has been received:
 - Check through the "Order Status Query" API if there has been a Pre-authorization granted. If there was a pre-authorization generated on the CMI side:
 - Action to take:
 - Call the "PostAuth" API to generate a Post authorization on the CMI side and debit the customer.
 - Update the database of the merchant website to record the payment information found on the CMI side.

4.2.2.Redirection back to merchant web site:

When a transaction is processed, a link is shown in the payment result page so that the customer can go back to the merchant's web site.

In the case of an approved transaction, the customer will be redirected to **okUrl** (parameter sent by the merchant in the payment request). All input parameters along with transaction response parameters will be posted to **okUrl**, and the "Response" parameter (see below) value will be **"Approved"**.

In the case of a failed transaction, the customer will be redirected to **failUrl** (parameter sent by the merchant in the payment request). All input parameters along with transaction response parameters will be posted to **failUrl**, and the "Response" parameter (see below) will be **"Declined"** or **"Error"**.

Notice:

- In general, the update of the payment status of the order at the merchant website is done at the time of the server-to-server callback request (via CallbackURL). It is recommended that the merchant website also checks the payment status of the order when the customer is redirected to okURL or failURL. Indeed, in the case where the request of the callback server-to-server (via the CallbackURL) fails, the redirection of the customer to the okURL or the failURL can be used to take into account the payment status that was processed in the CMI platform. We have just to remember that in the case of the server-to-server callback request (via the CallbackURL), the CMI platform expects a response from the merchant website to decide whether the customer should be debited or not; while in the case of customer redirection to okURL or failURL, no response is expected from the merchant website. So in this case the merchant must manually process the confirmation of the transaction via its Merchant Center CMI to debit the customer if necessary.

4.2.3. Payment result request's parameters:

When CMI platform sends back the payment result to the merchant web site (either via host-to-host callback or via customer redirection back), the request contains, in addition to the parameters sent by the merchant in the payment request, the following parameters:

Attribute name	Description	Format	Constraint
Response	Payment status	Alphabetic	Possible values: "Approved", "Error", "Declined"
ProcReturnCode	Transaction status code	Alphanumeric (2)	Possible values: "00" for authorized transactions, "99" for gateway errors, others for ISO-8583 error codes
EXTRA.TRXDAT E	Transaction Date	Alphanumeric (17)	Formatted as "dd/mm/yyyy hh24:mi:ss"
AuthCode	Transaction Verification/Approval/Authorization code	Alphanumeric (6)	
extra.cvvverification	Résultat de vérification de la banque du code CVV.	Alphanumeric (1)	Possible values: {M} The card verification code is valid. {N} The card verification code is not valid. {P} The code could not be verified. {S} Code not filled in. {U} Code not supported by the bank. {I} Invalid code. {X} No response from the bank.
ccnhash	Hash of the used payment card.	Alphanumeric (128)	Valeur unique par carte.
TDS2.dsTransID	Transaction ID in the card's DS (Directory Server).	Alphanumeric (50)	
orghash	HASH value that was sent by the merchant site to the CMI platform when the customer was redirected to the online payment page.	Alphanumeric	
orgrnd	RND value that was sent by the merchant site to the CMI platform when the customer was redirected to the online payment page.	Alphanumeric	
acqStan	A payment identifier (Payment ID) shared with acquirer and issuer	Numeric (6)	

HostRefNum	Host reference number	Alphanumeric (12)	
TransId	CMI platform Transaction Id	Alphanumeric (64)	
ErrMsg	Error message	Alphabetic (255)	
ClientIp	IP address of the customer	Alphanumeric (15)	Formatted as "###.###.###.###"
ReturnOid	Returned order ID	Alphanumeric (64)	Must be the same as input oid (sent in the payment request of the merchant)
paymentType	Used payment method	Alphanumeric	Possible values: CARD
EXTRA.CARDBRAND	Card brand name	Alphabetic	Possible values: VISA, MASTERCARD
MaskedPan	Masked credit card number	Alphanumeric (19)	Formatted as XXXXXX*****XXXX
cardHolderName	Card holder name used by the customer when processing the transaction	Alphanumeric	
Ecom_Payment_Card_ExpDate_Year	Card expiry year	Numeric (2)	
Ecom_Payment_Card_ExpDate_Month	Card expiry month	Numeric (2)	
EXTRA.CARDISSUER	Card issuer name	Alphabetic	
merchantID	MPI merchant ID	Alphanumeric (15)	
ACQBIN	Acquirer BIN	Numeric (6)	
mdStatus	Status code for the 3D transaction	Numeric (1)	Possible values: 1=authenticated transaction (Full 3D) 2, 3, 4 = Card not participating or attempt (Half 3D) 5,6,7,8 = Authentication not available or system error 0 = Authentication failed
txstatus	3D status for archival	Alphabetic (1)	Possible values "A", "N", "Y"
iReqCode	Code provided by ACS indicating data that is formatted correctly, but which invalidates the request. This element is included when business processing cannot be performed for some reason.	Numeric (2)	
iReqDetail	May identify the specific data elements that caused the Invalid Request Code (so never supplied if Invalid Request Code is omitted).	Alphanumeric	
vendorCode	Error message describing iReqDetail error.	Alphanumeric	
PAResSyntaxOK	If PARes validation is syntactically correct, the value is true. Otherwise value is false.	Alphabetic (1)	Possible values "N", "Y"
ParesVerified	If signature validation of the return message is successful, the value is true. If PARes message is not received or signature validation fails, the value is false.	Alphabetic (1)	Possible values "N", "Y"

eci	Electronic Commerce Indicator	Numeric (2)	Empty for non-3D transactions
cavv	Cardholder Authentication Verification Value, determined by ACS.	Alphanumeric (28)	Contains a 20 byte value that has been Base64 encoded, giving a 28 byte result.
xid	Unique internet transaction ID	Alphanumeric (28)	Base64 encoded
cavvAlgorithm	CAVV algorithm	Numeric (1)	Possible values "0", "1", "2", "3"
md	MPI data replacing card number	Alphanumeric	
Version	MPI version information	Alphanumeric (3)	Like "2.0"
sid	Schema ID	Numeric (1)	Possible values: "1" for Visa, "2" for Mastercard
MdErrorMsg	Error Message from MPI (if any)	Alphanumeric (512)	
rnd	Random string, will be used for hash comparison	Alphanumeric (20)	
HASH	Hash value	Alphanumeric	

4.2.4.Hash generation of the payment result

(Use the same technique of the payment request hash's generation – See above)

Notice:

- Some parameters sent from the server-to-server request (via the CallbackURL) are mandatory while others are optional, meaning that they may or may not exist in the request. It is important to use all the parameters received in order to calculate the hash.

4.2.5.Callback requests data examples

Successful transaction case:

Parameter	Value	Parameter	Value
ACQBIN	439218	INVOICENUMBER	
acqStan	56928	iReqCode	
amount	27.47	iReqDetail	
AuthCode	746579	itemnumber1	a1
Bill2Email	test@cmi.co.ma	itemnumber2	b1
BillToCity	Casablanca	itemnumber3	c1
BillToCompany	Billing Company	lang	en
BillToCountry	504	MaskedPan	400000***7190
BillToName	Bill John Doe	md	400000:BD64FA1BA77717 D21DA26554D2DE0 C9E65AB5D5872737AE720 F5D2484180228C: 4024:##600000004

BillToPostalCode	12345	mdErrorMsg	
BillToStateProv	mystate	mdStatus	1
BillToStreet1	Address line 1	merchantID	600000004
BillToStreet2	Address line 2	MERCHANTSURCHARGE	
BillToStreet3	Address line 3	noCallbackPaymentMethodList	
BillToTelVoice	123456	oid	sfgzzy4
CallbackResponse		okUrl	http://test.cmi.co.ma/ok
callbackUrl	http://test.cmi.co.ma/callback	PAResSyntaxOK	
cardHolderName	Cardholder name	PAResVerified	
cavv	AAABCRA0I1WFQDgnWDSX AAAAAAA=	paymentType	CARD
cavvAlgorithm		payResults.dsId	1
choix1	on	pMethod	
clientid	600000004	price1	2.00
clientIp	81.192.141.16	price2	3.95
currency	504	price3	3.50
CUSTOMERSISDN		ProcReturnCode	0
CUSTOMERSURCHARGE		productcode1	a2
CVVPresence	1	productcode2	b2
desc1	desc1	productcode3	c2
desc2	desc2	qty1	3
desc3	desc3	qty2	1
digest	digest	qty3	5
DIMCRITERIA1		RecurringFrequency	
DIMCRITERIA10		RecurringFrequencyUnit	
DIMCRITERIA2		RecurringPaymentNumber	
DIMCRITERIA3		refreshTime	10
DIMCRITERIA4		Response	Approved
DIMCRITERIA5		ReturnOid	sfgzzy4
DIMCRITERIA6		rnd	lBjJfQCTTrNRfMcNe111
DIMCRITERIA7		sessiontimeout	
DIMCRITERIA8		SettleId	1
DIMCRITERIA9		ShipToCity	Casablanca
dsId	1	ShipToCompany	Shipping Company
eci	5	ShipToCountry	504
Ecom_Payment_Card_ExpDate_Month	12	ShipToName	Ship John Doe
Ecom_Payment_Card_ExpDate_Year	17	ShipToPostalCode	12345
EDITABLEORDERITEM		ShipToStateProv	mystate
ErrMsg		ShipToStreet1	Address line 1
EXCHANGEAMOUNT	@@EXCHANGEAMOUNT@@	ShipToStreet2	Address line 2
EXCHANGE CURRENCY		ShipToStreet3	Address line 3

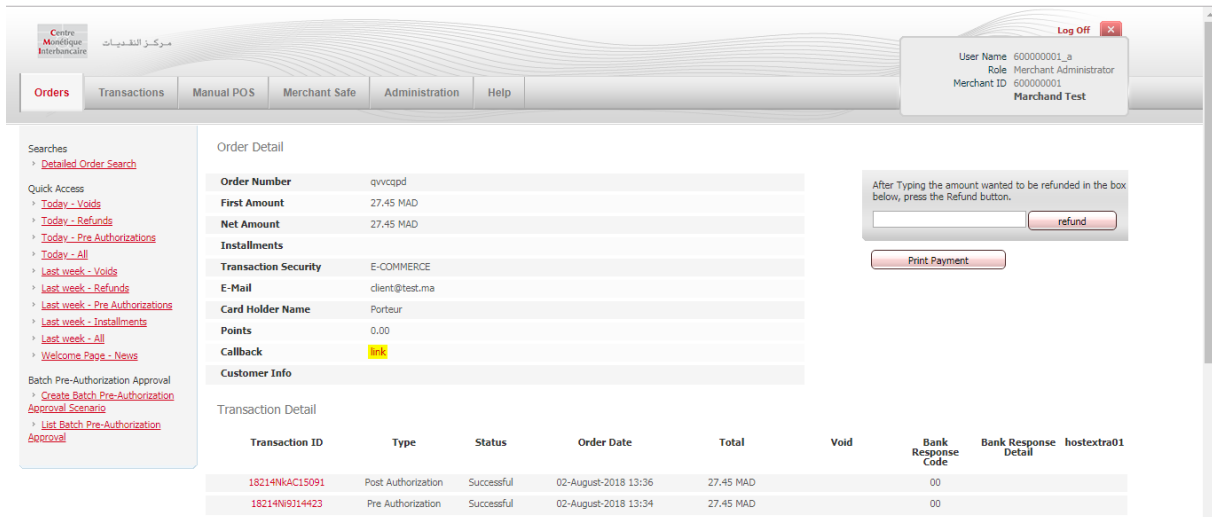
EXTRA.CARDBRAND	VISA	ShipToTelVoice	789456
EXTRA.CARDISSUER	CDM	shopurl	http://test.cmi.co.ma/shop
EXTRA.HOSTMSG	APPROVED	SID	
EXTRA.TRXDATE	23/11/02017 15:57:06	storetype	3D PAY HOSTING
failUrl	http://test.cmi.co.ma/fail	total1	6.00
fatouratiExpress	1	total2	3.95
GRACEPERIOD		total3	17.50
hashAlgorithm	ver3	TRANID	129430
HostRefNum	732715056928	TransId	17327P7GH13718
id1	id1	trantype	PreAuth
id2	id2	txstatus	Y
id3	id3	vendorCode	
instalment		version	
ccnhash	3M4jx5XsDuLOrICyvMP1ELPD +7pxU0A+8zSdx3FTkSA=	xid	7AFKUXDa5KeelWxM6wxfB9YfLDY= 62225a14-3945-4ece-9ed9- 336c478237ed
extra.cvvverification	M	TDS2.dsTransID	

Failed transaction case:

Parameter	Value	Parameter	Value
BillToCompany	Company	mdErrorMsg	Multiple values exist for the single parameter.
BillToName	Name	mdStatus	7
callbackUrl	http://test.cmi.co.ma/callback	oid	12345
clientid	600000000	okUrl	http://test.cmi.co.ma/ok
clientIp	1.1.1.1	price	
currency	504	ProcReturnCode	99
CVVPresence	1	productCode	
Ecom_Payment_Card_ExpDate_Month	1	qty	
Ecom_Payment_Card_ExpDate_Year	20	refreshtime	300
email	test@cmi.co.ma	ResponseError	
ErrCode	CORE-5110	retryXid	I6Hjx1K6zUGjqY98Z+vEnR2/gr4=
ErrMsg	Multiple values exist for the single parameter.	rnd	mmduZ3aMFe8qDmEG1MV1
failUrl	http://test.cmi.co.ma/fail	shopurl	http://test.cmi.co.ma/shop
hashAlgorithm	ver3	storetype	3d_pay_hosting
itemNumber		tel	600000000
lang	fr	total	
MaskedPan	400000***7190	trantype	PreAuth
		xid	I6Hjx1K6zUGjqY98Z+vEnR2/gr4=

4.2.6. Callbacks monitoring via CMI Merchant Center

The merchant has the possibility to view the details of the callback requests through his Merchant Center CMI where he is used to track his online transactions.



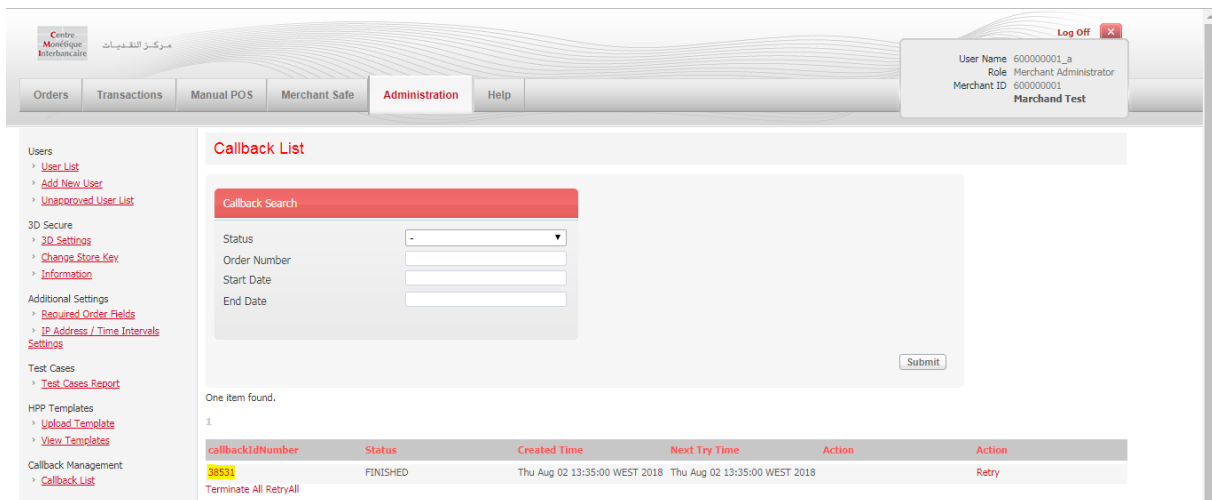
Order Detail

Order Number: qvvcqpd
 First Amount: 27.45 MAD
 Net Amount: 27.45 MAD
 Installments:
 Transaction Security: E-COMMERCE
 E-Mail: client@test.ma
 Card Holder Name: Porteur
 Points: 0.00
 Callback: [link](#)
 Customer Info

Transaction Detail

Transaction ID	Type	Status	Order Date	Total	Void	Bank Response Code	Bank Response Detail	hostextra01
18214N6AC15091	Post Authorization	Successful	02-August-2018 13:36	27.45 MAD		00		
18214N9314423	Pre Authorization	Successful	02-August-2018 13:34	27.45 MAD		00		

By clicking on the Callback link, in the order details, we can access the detail of the callback request that is exchanged between the CMI platform and the merchant website.



Callback List

Callback Search

Status:
 Order Number:
 Start Date:
 End Date:

Submit

One item found.

callbackIdNumber	Status	Created Time	Next Try Time	Action	Action
38531	FINISHED	Thu Aug 02 13:35:00 WEST 2018	Thu Aug 02 13:35:00 WEST 2018		Retry

We have to click on the callback's identifier to be able to access its detail.



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مركز المبيعات

Orders Transactions Manual POS Merchant Safe Administration Help

Log Off

User Name: 600000001_a
Role: Merchant Administrator
Merchant ID: 600000001
Marchand Test

Users
 > User List
 > Add New User
 > Unapproved User List

3D Secure
 > 3D Settings
 > Change Store Key
 > Information

Additional Settings
 > Required Order Fields
 > IP Address / Time Intervals
 Settings

Test Cases
 > Test Cases Report

HPP Templates
 > Upload Template
 > View Templates

Callback Management
 > Callback List

Merchant Terminal Information
 > Merchant Terminal Information

Payment Methods
 > Payment Methods

Payment Links
 > Payment Links

Callback List

Order Number	qvcqpd
id	38531
noCallbackPaymentMethodList	
Bill2Email	client@post.ma
clientid	600000001
Ecom_Payment_Card_ExpDate_Year	
qty3	5
qty2	1
BillToCompany	Billing Company
ShipToCompany	Shipping Company
qty1	3
BillToStreet3	
BillToStreet2	
BillToStreet1	Address
sessiontimeout	
BillToName	Client Name
CVVPresence	1
EXCHANGE CURRENCY	
ShipToStreet3	Address line 3
itemnumber3	c1
ShipToStreet2	Address line 2
itemnumber2	b1
itemnumber1	a1
dstid	0
choix1	on
DIMCRITERIA7	
ShipToTelVoice	789456
total1	6.00
DIMCRITERIA8	
total2	3.95
DIMCRITERIA9	
total3	17.50
ShipToStreet1	Address line 1
fatouratiExoress	1

We can then see all the data sent by the CMI platform to the merchant website in the callback request.

status	NO
created	Thu Aug 02 12:35:00 WEST 2018
dimuid	600000001
Callback Url	https://testpayment.cmi.co.ma/adapter/postauth.html
type	DIM_SYNC_CALLBACK
response	ACTION=POSTAUTH

We can even see the response of the merchant site to the request of the Callback of the CMI.

5. TRANSACTIONS' STATUSES

The payment process goes through several steps during which the transaction status changes. In this chapter, we detail this topic with illustrations from the CMI Merchant Center which is made available by CMI to merchants so that they can track their online transactions at any time.

5.1. PRE AUTHORIZATION :

When the customer enters his payment card details in the payment page, the CMI sends a request for payment authorization to the issuing bank. When the bank accepts the request, it blocks the transaction's amount in the customer's bank account. In this case, the customer is not charged. It is indeed a blocking of funds which guarantees the merchant the recovery of its funds if he accepts the transaction within the allowed time (up to 4 or 7 days).

At this stage, the transaction is processed within the Pre Authorization phase. This is visible through the Merchant Center CMI as shown below.



Centre Montépine Interbancaire

مرکز التجهيزات

Log Off

User Name: 600000001_a
Role: Merchant Administrator
Merchant ID: 600000001
Marchand Test

Orders Transactions Manual POS Merchant Safe Administration Help

Searches
Detailed Order Search

Quick Access
Today - Voids
Today - Refunds
Today - Pre Authorizations
Today - All
Last week - Voids
Last week - Refunds
Last week - Pre Authorizations
Last week - Installments
Last week - All
Welcome Page - News

Batch Pre-Authorization Approval
Create Batch Pre-Authorization Approval Scenario
Approval Scenario

Orders

One item found.

1

Refund	Approval	Order Number	Order Date	Name/Surname	Pos Entry Mode	Order Status	First Amount	Net Amount	Currency	Installments	Points	Transaction Security	Extra Info	Email
		sdypopz	02-August-2018 13:32	Client Name	E_COMMERCE	PRE	27.45	0.00	MAD		0.00			client@test.ma

Select All

Approve

Download to Excel

Currency: MAD Page Total: 0 Full List Total: 0

The order status is set to "PRE" which refers to the Pre Authorization.

Centre Montépine Interbancaire

مرکز التجهيزات

Log Off

User Name: 600000001_a
Role: Merchant Administrator
Merchant ID: 600000001
Marchand Test

Orders Transactions Manual POS Merchant Safe Administration Help

Searches
Detailed Order Search

Quick Access
Today - Voids
Today - Refunds
Today - Pre Authorizations
Today - All
Last week - Voids
Last week - Refunds
Last week - Pre Authorizations
Last week - Installments
Last week - All
Welcome Page - News

Batch Pre-Authorization Approval
Create Batch Pre-Authorization Approval Scenario
List Batch Pre-Authorization Approval

Order Detail

Order Number: sdypopz

First Amount: 27.45 MAD

Net Amount: 0.00 MAD

Installments

Transaction Security: E-COMMERCE

E-Mail: client@test.ma

Card Holder Name: Porteur

Points: 0.00

Customer Info

Transaction Detail

Transaction ID	Type	Status	Order Date	Total	Void	Bank Response Code	Bank Response Detail	hostextra01
18214NqoF13756	Pre Authorization	Successful	02-August-2018 13:32	27.45 MAD	Click To Void	00		

After entering the partial Post-Auth amount in the given box below, press the Close button.

Close

Print Payment

In the order details, we find a single transaction whose type is "Pre Authorization".

5.2. POST AUTHORIZATION :

When the CMI platform sends a callback request to the merchant website to return the payment result, the merchant website requests to debit the customer by replying with "ACTION=POSTAUTH". In this case, the transaction moves to the Post Authorization phase. This is visible through the Merchant Center CMI as shown below.



Centre Montépine Interbank

مرکز التجهيزات

Log Off

User Name: 600000001_a
Role: Merchant Administrator
Merchant ID: 600000001
Marchand Test

Orders Transactions Manual POS Merchant Safe Administration Help

Searches
Detailed Order Search

Quick Access
Today - Voids
Today - Refunds
Today - Pre Authorizations
Today - All
Last week - Voids
Last week - Refunds
Last week - Pre Authorizations
Last week - Installments
Last week - All
Welcome Page - News

Batch Pre-Authorization Approval
Create Batch Pre-Authorization Approval Scenario
List Batch Pre-Authorization Approval

Orders

One item found.

Refund	Approval	Order Number	Order Date	Name/Surname	Pos Entry Mode	Order Status	First Amount	Net Amount	Currency	Installments	Points	Transaction Security	Extra Info	Email
☐		qvcqpd	02-August-2018 13:34	Client Name	E_COMMERCE	POST	27.45	27.45	MAD		0.00			client@test.ma

Select All

Refund

Currency: MAD Page Total: 27.45 Full List Total: 27.45

Download to Excel

The order status is set to "POST" which refers to the Post Authorization.

Centre Montépine Interbank

مرکز التجهيزات

Log Off

User Name: 600000001_a
Role: Merchant Administrator
Merchant ID: 600000001
Marchand Test

Orders Transactions Manual POS Merchant Safe Administration Help

Searches
Detailed Order Search

Quick Access
Today - Voids
Today - Refunds
Today - Pre Authorizations
Today - All
Last week - Voids
Last week - Refunds
Last week - Pre Authorizations
Last week - Installments
Last week - All
Welcome Page - News

Batch Pre-Authorization Approval
Create Batch Pre-Authorization Approval Scenario
List Batch Pre-Authorization Approval

Order Detail

Order Number: qvcqpd
First Amount: 27.45 MAD
Net Amount: 27.45 MAD
Installments:
Transaction Security: E-COMMERCE
E-Mail: client@test.ma
Card Holder Name: Porteur
Points: 0.00
Callback: link
Customer Info:

After Typing the amount wanted to be refunded in the box below, press the Refund button.

refund

Print Payment

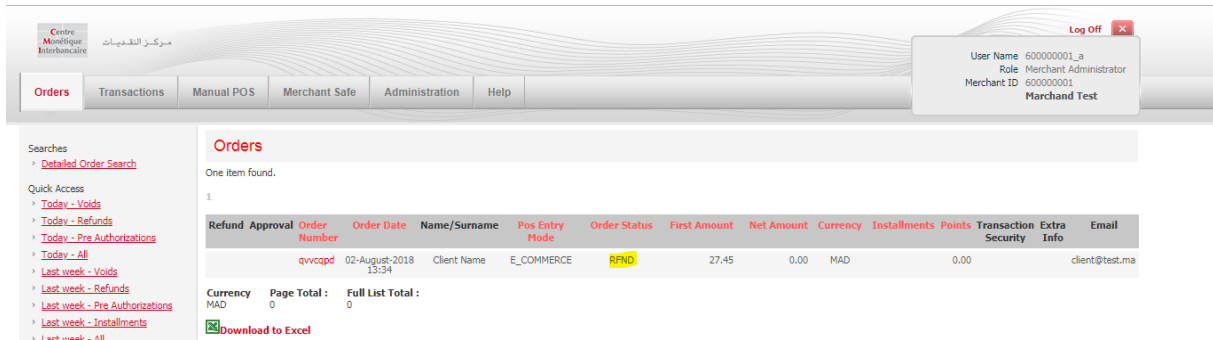
Transaction Detail

Transaction ID	Type	Status	Order Date	Total	Void	Bank Response Code	Bank Response Detail	hostextra01
18214NAC15091	Post Authorization	Successful	02-August-2018 13:36	27.45 MAD		00		
18214N9314423	Pre Authorization	Successful	02-August-2018 13:34	27.45 MAD		00		

In the order detail, there are two transactions. One of them is set to type "Pre Authorization" and the other one is set to type "Post Authorization".

5.3. REFUND :

Sometimes, the merchant wants to payback one of his customers that has processed an online transaction through the merchant's website. The merchant can in this case execute a refund operation through its CMI Merchant Center. Once executed, the transaction moves to the Refund phase. This is visible through the CMI Merchant Center as shown below.



Centre Montique Interbancaire

Log Off

User Name: 600000001_a
Role: Merchant Administrator
Merchant ID: 600000001
Marchand Test

Orders Transactions Manual POS Merchant Safe Administration Help

Searches
Detailed Order Search

Quick Access
Today - Voids
Today - Refunds
Today - Pre Authorizations
Today - All
Last week - Voids
Last week - Refunds
Last week - Pre Authorizations
Last week - Installments
All - Voids

Orders

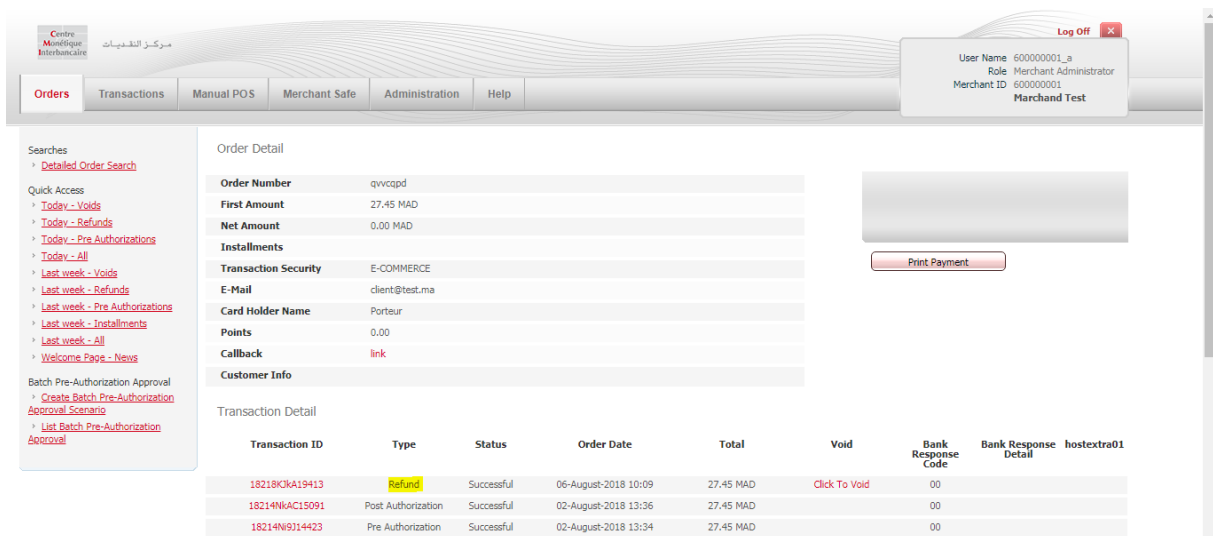
One item found.

Refund Approval Order Number	Order Date	Name/Surname	Pos Entry Mode	Order Status	First Amount	Net Amount	Currency	Installments	Points	Transaction Security	Extra Info	Email
qvcqpd	02-August-2018 13:34	Client Name	E_COMMERCE	RFND	27.45	0.00	MAD		0.00			client@test.ma

Currency: MAD Page Total: 0 Full List Total: 0

Download to Excel

The order status is set to "RFND" which refers to the Refund.



Centre Montique Interbancaire

Log Off

User Name: 600000001_a
Role: Merchant Administrator
Merchant ID: 600000001
Marchand Test

Orders Transactions Manual POS Merchant Safe Administration Help

Searches
Detailed Order Search

Quick Access
Today - Voids
Today - Refunds
Today - Pre Authorizations
Today - All
Last week - Voids
Last week - Refunds
Last week - Pre Authorizations
Last week - Installments
Last week - All
Welcome Page - News

Batch Pre-Authorization Approval
Create Batch Pre-Authorization Approval Scenario
List Batch Pre-Authorization Approval

Order Detail

Order Number: qvcqpd
First Amount: 27.45 MAD
Net Amount: 0.00 MAD
Installments: 0
Transaction Security: E-COMMERCE
E-Mail: client@test.ma
Card Holder Name: Porteur
Points: 0.00
Callback: link
Customer Info: link

Print Payment

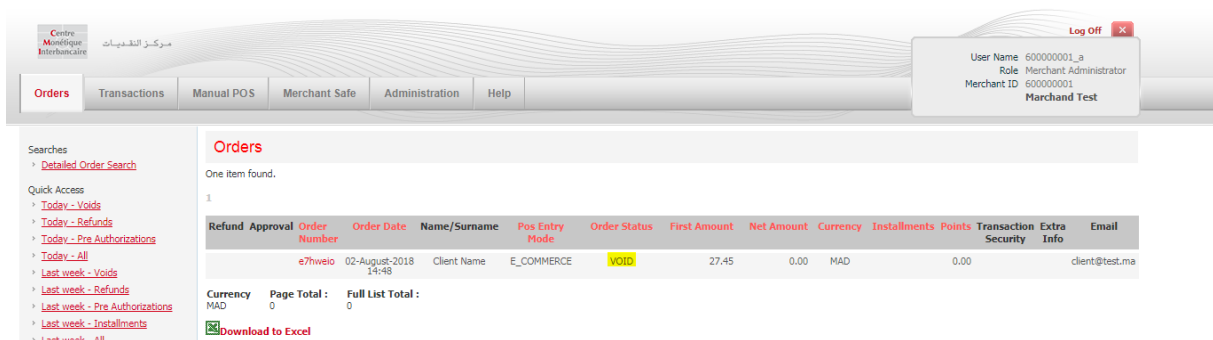
Transaction Detail

Transaction ID	Type	Status	Order Date	Total	Void	Bank Response Code	Bank Response Detail	hostextra01
18218KJkA19413	Refund	Successful	06-August-2018 10:09	27.45 MAD	Click To Void	00		
18214NkAC15091	Post Authorization	Successful	02-August-2018 13:36	27.45 MAD		00		
18214N9J14423	Pre Authorization	Successful	02-August-2018 13:34	27.45 MAD		00		

In the order detail, there are three transactions. One of them is set to type "Refund" and the other ones are set to types "Pre Authorization" and "Post Authorization".

5.4. VOID :

When the transaction is in the "Pre Authorization" phase, the merchant can execute its cancellation to request the release of funds at the customer's bank account. The cancellation operation can be done through the CMI Merchant Center. This is visible through the Merchant Center CMI as shown below.



Centre Montique Interbancaire

Log Off

User Name: 600000001_a
Role: Merchant Administrator
Merchant ID: 600000001
Marchand Test

Orders Transactions Manual POS Merchant Safe Administration Help

Searches
Detailed Order Search

Quick Access
Today - Voids
Today - Refunds
Today - Pre Authorizations
Today - All
Last week - Voids
Last week - Refunds
Last week - Pre Authorizations
Last week - Installments
All - Voids

Orders

One item found.

Refund Approval Order Number	Order Date	Name/Surname	Pos Entry Mode	Order Status	First Amount	Net Amount	Currency	Installments	Points	Transaction Security	Extra Info	Email
e7hweio	02-August-2018 14:48	Client Name	E_COMMERCE	VOID	27.45	0.00	MAD		0.00			client@test.ma

Currency: MAD Page Total: 0 Full List Total: 0

Download to Excel

The order status is set to "VOID" which refers to the Cancellation.



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مركز التجديدات

Log Off

User Name: 600000001_a
Role: Merchant Administrator
Merchant ID: 600000001
Marchand Test

OrdersTransactionsManual POSMerchant SafeAdministrationHelp

Searches
Detailed Order Search

Quick Access
Today - Voids
Today - Refunds
Today - Pre Authorizations
Today - All
Last week - Voids
Last week - Refunds
Last week - Pre Authorizations
Last week - Installments
Last week - All
Welcome Page - News

Batch Pre-Authorization Approval
Create Batch Pre-Authorization Approval Scenario
List Batch Pre-Authorization Approval

Order Detail

Order Number: e7hveio
First Amount: 27.45 MAD
Net Amount: 0.00 MAD
Installments
Transaction Security: E-COMMERCE
E-Mail: client@test.ma
Card Holder Name: Porteur
Points: 0.00
Callback: link
Customer Info

Print Payment

Transaction Detail

Transaction ID	Type	Status	Order Date	Total	Void	Bank Response Code	Bank Response Detail	hostextra01
18214OwyF19029	Pre Authorization	Void	02-August-2018 14:48	27.45 MAD		00		

In the order detail, the transaction's type is "Pre Authorization" with the status "Void".

Notice:

- For more information on Merchant Center CMI features, please refer to the appropriate user manual.

6. ANNEX

6.1. CMI ECOM Integration service's contact :

If you have a question about the technical interfacing of your web site with CMI online payment platform, you can contact our integration service via the following email:
integration.ecom@cmi.co.ma

6.2. Error codes :

The following table lists the main errors codes that identify transactions' failures and that can be found in the CMI Merchant Center with all other transactions' details:

Error Code	System Error Message	Err Desc EN	Err Desc FR	Err Desc AR	Comments
99	3D-1004	Wrong security code	Code de sécurité erroné	رمز الحماية خاطئ	Rejection of the platform due to hash code calculation error. This can happen either in the payment request, or in the payment result request.
99	3D-1005	Failed operation	Echec de l'opération	فشل العملية	Rejection of the platform due to cardholder online authentication failure.
99	3D-1034	Failed operation	Echec de l'opération	فشل العملية	Rejection of the platform due to mishandling of the client.
99	BM-1002	HOST based messaging problem	HOST based messaging problem	ملكش HOST	Invalid response message from acquirer.
99	BM-9101	Failed operation	Opération échouée	عملية فاشلة	Acquiring interface timeout.
99	BM-9102	Failed operation	Opération échouée	عملية فاشلة	Acquiring interface timeout.
99	CORE-2010	The credit card is expired.	La carte de crédit est expirée	عدم تجديد بطاقة ائتمان	Rejection of the platform due to expired credit card use.
99	CORE-2012	The credit card number is not in a valid format.	Le format du numéro de carte de crédit n'est pas valide.	رقم بطاقة الائتمان ليس في شكل صحيح	Rejection of the platform.
99	CORE-2202	Failed operation	Opération échouée	عملية فاشلة	Rejection of the platform due to a control rule.

					Example of control rules: Authorization request in case of 3D Secure authentication fallback.
99	CORE-2208	Card brand is not allowed.	Marque de carte non autorisée.	نوع بطاقة الانتماء غير مقبول	Rejection of the platform.
99	CORE-2253	The payment authorization could not be performed	Impossible de procéder à l'autorisation de paiement	لا يمكن إجراء عملية الدفع	Rejection of the platform due to a control rule. Example of control rules: Using a non-authenticatable 3D Secure foreign payment card.
99	CORE-2515	Incorrect data	Données incorrectes	بيانات غير صحيحة	Rejection of the platform due to bad data entered by the customer (Exp. Bad CVV Format).
99	CORE-5110	Please return to the web site and try again	Merci de retourner au site web et réessayer	يرجى العودة إلى الموقع الإلكتروني وإعادة المحاولة	Rejection of the platform due to mishandling of the client.
99	RULE-0001	Your session has expired. Please return to the merchant website and try again.	Votre session a expiré. Merci de revenir au site marchand et réessayer.	انتهى الوقت المسموح. المرجو العودة إلى موقع التاجر والمحاولة مرة أخرى	CMI platform rejection due to session timeout.
99	RULE-0002	Cardholder authentication required	Authentification du porteur requise	مصادقة حامل البطاقة مطلوبة	CMI platform rejection due to cardholder authentication lack.
99	RULE-0003	Wrong card type	Type de carte erroné	نوع البطاقة خاطئ	CMI platform rejection due to wrong card type use.
99	RULE-0004	Card not allowed	Carte non permise	البطاقة غير مسموح بها	CMI platform rejection due to not allowed card use.
99	CORE-2508	Cannot refund, transaction not settled.	Vous ne pouvez pas rembourser, transaction non réglée.	لا يمكن برد، الصفقة لم تحسم.	CMI platform rejection. It is not possible to send a refund transaction until settlement.
99	CORE-1004	System error. Response parameters general exception.	Erreur système. Paramètres de réponse d'exception générale.	خطأ في النظام. المعلومات استجابة استثناء عام.	CMI platform rejection. System error occurred.
03	ISO8583-03	Payment authorization not permitted for this merchant	Autorisation de paiement non permise pour ce marchand	عملية الدفع غير مسموح بها لهذا التاجر	Bank rejection.
04	ISO8583-04	Payment authorization rejected with this card	Autorisation de paiement rejetée avec cette carte	تم رفض عملية الدفع بهذه البطاقة	Rejection of the bank for stolen credit card use.

05	ISO8583-05	Payment authorization declined	Autorisation de paiement non acceptée	تم رفض عملية الدفع	Rejection of the bank without precision of the reason.
12	ISO8583-12	Payment authorization rejected	Autorisation de paiement rejetée	تم رفض عملية الدفع	Bank rejection.
13	ISO8583-13	INVALID AMOUNT	Montant non valide	غالباً ريغ جلاص	Bank rejection.
14	ISO8583-14	Payment authorization rejected with the used card	Autorisation de paiement rejetée avec la carte utilisée	تم رفض عملية الدفع بالبطاقة المستعملة	Bank rejection.
15	ISO8583-15	Payment authorization failed	Autorisation de paiement échouée	فشل عملية الدفع	Bank rejection.
39	ISO8583-39	No credit account	Pas de compte de crédit	سلا كريدل باسحدا ممتلا	Bank rejection.
51	ISO8583-51	Insufficient funds.	Solde de la carte insuffisant	رصيد غير كاف	Bank rejection.
54	ISO8583-54	Payment authorization rejected with this card	Autorisation de paiement rejetée avec cette carte	تم رفض عملية الدفع بهذه البطاقة	Rejection of the bank due to expired card use or incorrect card expiry date use.
57	ISO8583-57	Payment authorization not permitted	Autorisation de paiement non permise	عملية الدفع غير مسموح بها	Bank rejection.
61	ISO8583-61	Activity amount limit exceeded.	Plafond dépassé	تجاوز السقف	Bank rejection.
62	ISO8583-62	Payment authorization rejected with this card	Autorisation de paiement rejetée avec cette carte	تم رفض عملية الدفع بهذه البطاقة	Bank rejection.
63	ISO8583-63	Payment authorization rejected with this card	Autorisation de paiement rejetée avec cette carte	تم رفض عملية الدفع بهذه البطاقة	Rejection of the bank due to security reason.
65	ISO8583-65	Activity limit exceeded.	Plafond dépassé	تجاوز السقف	Bank rejection.
82	ISO8583-82	CVV Failure or CVV Value supplied is not valid.	Echec CVV ou valeur CVV fournie non valide.	رمز التحقق غير صحيح	Bank rejection.
86	ISO8583-86	Payment authorization rejected	Autorisation de paiement rejetée	تم رفض عملية الدفع	Bank rejection.
90	ISO8583-90	Payment authorization rejected	Autorisation de paiement rejetée	تم رفض عملية الدفع	Bank rejection.
91	ISO8583-91	Payment authorization failed	Autorisation de paiement échouée	فشل عملية الدفع	Bank interface timeout.
96	ISO8583-96	Payment authorization failed	Autorisation de paiement échouée	فشل عملية الدفع	Failed operation.
06	ISO8583-06	Error (found only in file update responses).	Erreur (trouvé seulement dans le fichier des mise à jour des réponses).	خطأ (وجدت فقط في الردود تحديث الملف).	Bank rejection.
01	ISO8583-01	Referral - call bank for manual approval.	Referall-appeler la banque pour approbation manuelle.	الإحالة - دعوة البنك للموافقة اليدوية.	Bank rejection.
55	ISO8583-55	Incorrect PIN	PIN incorrecte	PIN غير صحيح.	Bank rejection.

75	ISO8583-75	Allowable number of PIN tries exceeded.	Nombre de tentatives PIN autorisé dépassée.	عدد المسموح به من محاولات PIN تجاوز.	Bank rejection.
80	ISO8583-80	Date is not valid.	La date est non valide	تاريخ غير صالح.	Bank rejection.
93	ISO8583-93	Violation, cannot complete (installment, loyalty).	Violation, cannot complete (installment, loyalty	انتهاك، لا يمكن إكمال (القسط، ولاء).	Bank rejection.

Notice:

- Web developers often encounter the 3D-1004 error during their first integration tests. In this case, the origin of the error usually comes from the fact that the developer forgets to enter his secret hash key at Merchant Center CMI. The procedure for setting the secret hash key is specified in the CMI integration kit.